

New Tinnitus Questionnaire Is a Sensitive Tool for Clinical Trials

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The psychological model of tinnitus proposed by Dauman and Tyler distinguished tinnitus from the reaction to it.¹ Several counseling and sound therapy approaches to these reactions have evolved from the pioneering work of Hallam's habituation model² and from the cognitive behavioral strategies suggested by Sweetow³ and by Henry and Wilson.⁴ (For a review of these strategies, see reference 5.)

Our own protocol, Tinnitus Activities Treatment, includes cognitive behavioral modification principles, acceptance, and attention-control exercises.⁶⁻⁷

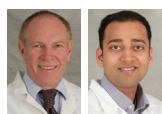
Development and Validation of the Tinnitus Primary Function Questionnaire

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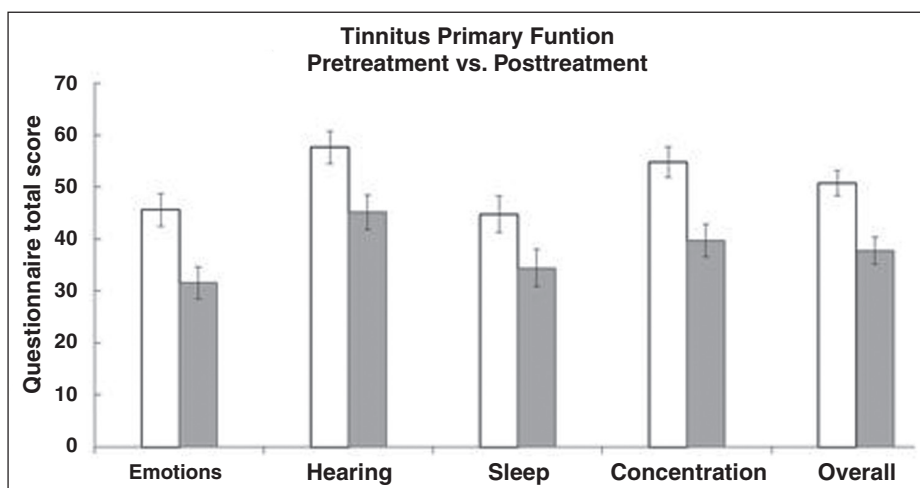
We developed a questionnaire that focuses on how tinnitus affects a person's life. Our recent publication documents the validity, reliability, and sensitivity of the new Tinnitus Primary Function Questionnaire.

The questionnaire evolved from our clinical experience with the four primary areas affected by tinnitus: hearing, concentration, thoughts and emotions, and sleep. These primary functions can affect secondary factors such as work, recreation, friendships, and other general quality-of-life issues.

We analyzed data from 158 patients who had received an early version of the Tinnitus Activities Treatment. A factor analysis confirmed the validity of the four main functions affected by tinnitus.



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Overall and subscale scores on the Tinnitus Primary Function Questionnaire demonstrated improvement with treatment. The white bars show pretreatment scores, and the gray bars show posttreatment scores. (Modified from Tyler R, Ji H, Perreau A, Witt S, Noble W, Coelho C. Development and validation of the Tinnitus Primary Function Questionnaire. *Am J Audiol* 2014;23[3]:260-272. Available at: bit.ly/AJA-Tinnitus.)

The Tinnitus Handicap Questionnaire,⁸ which is employed worldwide to evaluate treatments, was used as a metric of validity in our study, as were the Beck Depression Inventory (BDI),⁹ the State-Trait Anxiety Questionnaire,¹⁰ and the Pittsburgh Sleep Quality Index (PSQI).¹¹

"The questionnaire evolved from our clinical experience with the four primary areas affected by tinnitus: hearing, concentration, thoughts and emotions, and sleep."

All four questionnaires were correlated with the four subscales of the Tinnitus Primary Function Questionnaire. Cronbach's α , which measures inter-item correlation, indicated high internal consistency for the new questionnaire.

Scores on the Tinnitus Primary Function Questionnaire responded to treatment, as shown in the bar graph.

FOCUS ON PRIMARY FUNCTIONS

Two advantages of the Tinnitus Primary Function Questionnaire are its zero-to-100 measurement scale and focus on

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primary functions, excluding secondary functions such as quality of life.

Since a questionnaire should respond to treatment changes, it is important to use a measurement scale with ample resolution. Many questionnaires over the years have used interval scales from zero to 100, like the Tinnitus Primary Function Questionnaire does. When responding, patients typically choose a score of 0, 5, 10, 15, etc., resulting in 21 steps. This provides greater resolution than a zero-to-10 scale.

General quality-of-life questions can be influenced by family, social, and work issues. Therefore, their inclusion is likely to decrease a tinnitus questionnaire's responsiveness. A treatment might be effective, but, because of other life issues involving work or family, the questionnaire becomes a less sensitive tool.

Another unique attribute of the Tinnitus Primary Function Questionnaire is that the responsiveness to treatments can be even further improved by removing subscales for which a patient has no problems. For example, if a patient has no trouble sleeping, then all sleep questions will be insensitive to the treatment.

Before a trial begins, researchers can choose a within-subject design and plan to eliminate subscales with low handicap scores. Such a-priori screening to select the functions that are most likely to show changes can further improve the questionnaire's sensitivity.¹²

Many tinnitus questionnaires have been successfully used to document tinnitus distress and measure the effectiveness of treatments. Newman, Wharton, and Jacobson recommended the Tinnitus Handicap Questionnaire for clinical trials.^{8,13} Future work is needed to compare the validity, reliability, and responsiveness of the Tinnitus Primary Function Questionnaire

with other questionnaires such as the Tinnitus Handicap Questionnaire.

The 12-item version of the Tinnitus Primary Function Questionnaire, which also has been shown to be valid and reliable, can be administered quickly in the clinic, thus helping save precious time. ■